Find $\sup(A)$, $\max(A)$, $\inf(A)$, $\min(A)$

① $n \neq 0$

a) $A = \left\{ \frac{1}{n}, n \in \mathbb{N}^* \right\} = \left\{ 1, \frac{1}{2}, \frac{1}{3}, \dots \right\}$

$ub(A) = [1, \infty) \Rightarrow \sup(A) = 1 \in A \Rightarrow \exists \max A = 1.$

$lb(A) = (-\infty, 0] \Rightarrow \inf(A) = 0 \notin A \Rightarrow \nexists \min A.$

? $1 \overset{?}{\in} lb(A)$, $1 \leq a$, $(\forall) a \in A$ False.

? Let $x > 0$. Then $\exists n \in \mathbb{N}$ s.t. $x > \frac{1}{n}$, $n > \frac{1}{x}$, $n = \left[\frac{1}{x}\right] + 1.$

$\boxed{\min(A) = lb(A) \cap A}$ $= \emptyset$ (here)

b) $A = \{ x \in \mathbb{Q} \mid x^2 \leq 2 \} = [-\sqrt{2}, \sqrt{2}] \cap \mathbb{Q}$

$ub(A) = [\sqrt{2}, \infty)$

$lb(A) = (-\infty, -\sqrt{2}]$

$\sup(A) = \sqrt{2} \notin A \Rightarrow \nexists \max$

$\inf(A) = -\sqrt{2} \notin A \Rightarrow \nexists \min$

! Every set that is bounded above has a sup.

! " ——— " ——— an inf.

THEOREM.

$A \subseteq \mathbb{R}$ bounded set.

(1) $(\forall)\, \varepsilon > 0$, $\exists x \in A$ s.t. $\sup A - \varepsilon < x$ (P_1)

(2) $(\forall)\, \varepsilon > 0$, $\exists x \in A$ s.t. $x < \inf(A) + \varepsilon$ (P_2)

Proving (1):

Let $\varepsilon > 0$. Assume that $(\forall) x \in A$, $\sup A - \varepsilon \geq x$ $(!P_1)$

$\Rightarrow \sup A - \varepsilon \in ub A$

and

$\sup A - \varepsilon < \sup A \quad (*)$

But $\sup A$ is the smallest ub:

$(\forall) u \in ub(A),\ u \geq \sup(A) \quad (**)$

$(*), (**) \Rightarrow$ CONTRADICTION. $\Rightarrow P_1$ is true.

<u>Proving (2):</u>

Let $\varepsilon > 0$. Assume that $\underline{(\forall) x \in A : x \geq \inf(A) + \varepsilon}$ $(!P_2)$

$\Rightarrow \inf(A) + \varepsilon \in lb(A)$

and

$\inf A + \varepsilon > \inf \quad (*)$

But $\inf A$ is the largest lb:

$(\forall) u \in lb(A),\ u \leq \inf A \quad (**)$

$(*), (**) \Rightarrow$ CONTRADICTION $\Rightarrow P_2$ is true.

! Bounded set $\rightarrow$ always $\inf \leq \sup$
with equality when we have only one element in the set.
(Ex: $A = \{5\} \Rightarrow \inf A = \sup A = 5$)

<u>PROPOSITION</u>

$A \subseteq B \subseteq \mathbb{R}$ nonempty bounded sets.

$\inf B \leq \inf A \leq \sup A \leq \sup B$

Ex: $A = (2, 3)$ — $\inf A = 2$, $\sup A = 3$

$B = [1, 5)$ — $\inf B = 1$, $\sup B = 5$

$1 \leq 2 \leq 3 \leq 5$

$\sup(A \cup B) = \max\{\sup A, \sup B\}$

$\inf(A \cup B) = \min\{\inf A, \inf B\}$

$\overline{\mathbb{R}} := \mathbb{R} \cup \{-\infty, \infty\}$

↳ If A is <u>not</u> bounded above : $\sup A = \infty$

↳ If A is <u>not</u> bounded below: $\inf A = -\infty$

! $\emptyset$ is bounded by any number.

$\sup(\emptyset) = -\infty$

$\inf(\emptyset) = \infty$

<u>Neighbourhoods</u>:

$V \subseteq \mathbb{R}$ is a neighbourhood of $x \in \mathbb{R}$ if $\exists\, \varepsilon > 0$ s.t. $(x - \varepsilon, x + \varepsilon) \subseteq V$

- neighbourhood of ∞ : $\exists\, a \in \mathbb{R}$ s.t. $(a, \infty) \subseteq V$.
- neighbourhood of $-\infty$: $\exists\, a \in \mathbb{R}$ s.t. $(-\infty, a) \subseteq V$.

$\mathcal{V}(x) = \{V \subseteq \mathbb{R} \mid V \text{ neighbourhood of } x\}$

<u>Interior</u>:

$\operatorname{int}(A) = \{x \in \mathbb{R} \mid \exists\, V \in \mathcal{V}(x) \text{ s.t. } V \subseteq A\}$

Ex: $A = [0, 1)$

$\operatorname{int}(A) = (0, 1)$

Closure :

$$cl(A) = \{x \in \mathbb{R} \mid \exists V \in \mathcal{V}(x) \text{ s.t. } V \cap A \neq \emptyset\}$$

Ex: $[0,1) = A$

$cl(A) = [0,1]$.